

Notes: McLean & Pontiff (JF, 2016)

Does Academic Research Destroy Stock Return Predictability?

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Predictor sample	2
2.2	Portfolio construction	3
2.3	Period definitions	3
2.4	Publication date	3
2.5	Summary statistics	3
2.6	Arbitrage-cost and trading-activity measures	4
3	Empirical specifications and interpretation	4
3.1	Baseline decay regression	4
3.2	Three benchmark hypotheses	4
3.3	Interaction with in-sample performance	5
3.4	Finer event-time dynamics	5
3.5	Time-trend and persistence regressions	5
3.6	Predictor-type regressions	5
3.7	Limits-to-arbitrage regressions	6
3.8	Trading-activity dynamics	6
3.9	Comovement with other published predictors	6
4	Identification and empirical design	6
4.1	What identifies data mining versus publication effects	6
4.2	Why the post-sample coefficient is an upper bound on statistical bias	7
4.3	Why the post-publication coefficient is informative	7
4.4	Standard errors and cross-predictor dependence	7
4.5	Main limitations	7
5	Tables and figures	8
5.1	Figure 1 + Figure 2: strongest predictors decay more and returns decline around sample/publication dates	8
5.2	Table I + Table II: summary statistics and baseline return decay	9
5.3	Table III + Table IV: time trends, persistence, and predictor types	10
5.4	Table V + Table VI: arbitrage costs and trading activity	10
5.5	Table VII: comovement with other predictor indices	11
6	Short summary	12
7	Conclusion	12
7.1	Core conclusion	12
7.2	Economic intuition	12
7.3	Takeaway	12

1 Big picture

- **Question.** What happens to published stock-return predictors after the original sample period ends and after the academic paper is published?
- **Sample.** The authors study 97 cross-sectional stock-return predictors from 80 peer-reviewed finance, accounting, and economics papers.
- **Three periods.** For each predictor, they compare returns in the original in-sample period, the out-of-sample but pre-publication period, and the post-publication period.
- **Core result.** Average long-short predictor returns are 26% lower out of sample and 58% lower after publication.
- **Interpretation of out-of-sample decline.** The out-of-sample decline is interpreted as an upper-bound estimate of statistical bias and data mining, because sophisticated investors may already learn before publication.
- **Interpretation of post-publication decline.** The additional decline after publication is interpreted as evidence that publication informs investors, who then trade against the mispricing.
- **Publication effect.** The paper estimates a lower bound publication-informed trading effect of roughly 32%, computed as 58% post-publication decay minus 26% post-sample decay.
- **Persistence.** Predictability does not disappear completely. Post-publication average returns are lower but still positive.
- **Cross-sectional mechanism.** Predictors with higher in-sample returns and higher in-sample t -statistics experience larger declines after publication.
- **Arbitrage limits.** Post-publication returns are higher for predictors concentrated in stocks that are harder to arbitrage, such as stocks with higher idiosyncratic risk or lower liquidity.
- **Trading evidence.** After publication, trading volume, dollar volume, and short interest in predictor-portfolio stocks increase.
- **Comovement evidence.** After publication, each predictor's return becomes more correlated with already published predictors and less correlated with not-yet-published predictors.
- **Economic takeaway.** Academic research does not mechanically destroy every anomaly, but it appears to accelerate investor learning and arbitrage, causing substantial decay in return predictability.

2 Data and measurement

2.1 Predictor sample

- The paper collects published variables shown to predict cross-sectional stock returns.
- It only includes peer-reviewed studies in finance, accounting, and economics journals.
- It requires that the original paper reject no predictability at the 5% level.
- The predictor must be constructible from publicly available data.
- The final sample contains 97 predictors from 80 papers.
- The sample period ends in 2013.

Interpretation: The key unit of observation is not a firm; it is a predictor-month return. Each predictor is treated as a trading strategy whose return can be tracked before and after publication.

2.2 Portfolio construction

- For each predictor, the authors form a long-short portfolio.
- The portfolio buys one extreme quintile and shorts the opposite extreme quintile.
- The long side is chosen to match the sign of the original paper’s documented predictability.
- Returns are equal-weighted unless the original paper’s primary result uses value weighting.
- Indicator predictors are handled by comparing indicated firms with the complement set.
- Some original predictors cannot be replicated exactly because data changed or original details are incomplete; the authors construct close proxies in those cases.

Interpretation: The goal is not perfect replication of each study. The goal is a common return series that captures each published predictor’s economic content.

2.3 Period definitions

For each predictor i and month t , the sample is split into three parts:

- **In-sample:** months included in the original paper’s sample period.
- **Post-sample, pre-publication:** months after the original sample end but before the official publication date.
- **Post-publication:** months after the official publication date.

Interpretation: The post-sample period is used to detect data-mining or statistical bias. The post-publication period combines statistical-bias effects with investor learning from the published paper.

2.4 Publication date

- The baseline publication date is the month on the journal cover.
- Alternative definitions based on library arrival dates or SSRN dates produce similar results.
- Publication is only a proxy for investor awareness: some traders may learn before publication, and others may learn years later.

Interpretation: Because some investors learn before formal publication, the post-sample coefficient may already contain some publication-informed trading. This is why the paper calls the out-of-sample decline an upper bound on statistical bias and the publication effect a lower bound.

2.5 Summary statistics

- The average predictor has a monthly in-sample return of 0.582%.
- The average out-of-sample but pre-publication return is 0.402%.
- The average post-publication return is 0.264%.
- Average publication year is 2000 and median publication year is 2001.
- The average in-sample period is 323 months, the average post-sample pre-publication period is 56 months, and the average post-publication period is 156 months.

Interpretation: The raw averages already show the paper’s main result: returns fall after the original sample ends and fall further after publication.

2.6 Arbitrage-cost and trading-activity measures

The paper studies whether decay is related to limits to arbitrage using portfolio-level averages of:

- size;
- bid-ask spreads;
- dollar volume;
- idiosyncratic risk;
- dividend-paying status;
- a principal component index of the above variables.

It also studies trading activity through:

- stock-return variance;
- trading volume;
- dollar volume;
- short interest difference between the short and long sides of the predictor portfolio.

Interpretation: If publication induces arbitrage, then decay should be larger where arbitrage is cheaper, and trading activity should increase after publication.

3 Empirical specifications and interpretation

3.1 Baseline decay regression

The baseline panel regression is

$$R_{i,t} = \alpha_i + \beta_1 \text{PostSample}_{i,t} + \beta_2 \text{PostPublication}_{i,t} + e_{i,t}. \quad (1)$$

- $R_{i,t}$ is the monthly long-short return of predictor i in month t .
- α_i are predictor fixed effects.
- $\text{PostSample}_{i,t}$ equals one after the original sample ends but before publication.
- $\text{PostPublication}_{i,t}$ equals one after publication.

Interpretation: β_1 measures return decay outside the original sample and is interpreted as an upper bound on statistical bias. β_2 measures total post-publication decay, combining statistical bias and publication-induced investor learning.

3.2 Three benchmark hypotheses

The paper uses the baseline regression to distinguish three views:

- **Pure statistical bias:** both β_1 and β_2 should be approximately the negative of the in-sample mean, -0.582 .
- **Pure mispricing corrected by publication:** β_1 should be near zero and β_2 should be close to -0.582 .
- **No bias and no publication effect:** both coefficients should be zero.

Interpretation: The data reject all three extreme views. Predictability decays out of sample, decays further after publication, and remains positive after publication.

3.3 Interaction with in-sample performance

To test whether stronger published predictors decay more, the authors estimate:

$$R_{i,t} = \alpha_i + \beta_1 \text{PostSample}_{i,t} + \beta_2 \text{PostPublication}_{i,t} + \gamma_1 (\text{PostSample}_{i,t} \times \text{Mean}_i) + \gamma_2 (\text{PostPublication}_{i,t} \times \text{Mean}_i) + e_{i,t}. \quad (2)$$

They also replace Mean_i with the in-sample t -statistic:

$$R_{i,t} = \alpha_i + \beta_1 \text{PostSample}_{i,t} + \beta_2 \text{PostPublication}_{i,t} + \delta_1 (\text{PostSample}_{i,t} \times t\text{-stat}_i) + \delta_2 (\text{PostPublication}_{i,t} \times t\text{-stat}_i) + e_{i,t}. \quad (3)$$

Interpretation: Larger in-sample returns and larger in-sample t -statistics predict larger post-publication declines. This supports both data-mining concerns and the idea that arbitrageurs target the most profitable predictors.

3.4 Finer event-time dynamics

Figure 2 and related regressions split the post-sample and post-publication periods more finely:

- last year in sample;
- first year out of sample;
- later out-of-sample but pre-publication months;
- first through fifth years after publication;
- years beyond five after publication.

Interpretation: Returns are strong in the last in-sample year, decline outside the original sample, and decline more after publication. The gradual post-publication pattern is consistent with slow investor learning and slow arbitrage capital deployment.

3.5 Time-trend and persistence regressions

The paper tests whether the results are explained by:

- a general time trend in return predictability;
- a post-1993 break;
- month fixed effects;
- persistence in predictor returns using one-month and twelve-month lagged returns.

Interpretation: The post-publication decline remains robust to these controls, so the main result is not merely a general secular decline in anomaly returns.

3.6 Predictor-type regressions

Predictors are grouped into:

- market variables;
- event variables;
- valuation variables;
- fundamental variables.

The generic regression is

$$R_{i,t} = \alpha + \beta \text{PostPublication}_{i,t} + \theta \text{Type}_i + \phi(\text{PostPublication}_{i,t} \times \text{Type}_i) + e_{i,t}. \quad (4)$$

Interpretation: Predictors based on market information, such as price and trading data, decay more after publication than predictors based on less easily traded information. This is consistent with publication-induced arbitrage.

3.7 Limits-to-arbitrage regressions

The costly-arbitrage regressions estimate:

$$R_{i,t} = \alpha_i + \beta \text{PostPublication}_{i,t} + \theta CA_{i,t} + \phi(\text{PostPublication}_{i,t} \times CA_{i,t}) + e_{i,t}, \quad (5)$$

where $CA_{i,t}$ is a costly-arbitrage proxy such as size, spreads, dollar volume, idiosyncratic risk, dividends, or an index.

Interpretation: Returns persist more when arbitrage is difficult. This helps explain why publication reduces but does not fully eliminate predictability.

3.8 Trading-activity dynamics

For each trading-activity characteristic $X_{i,t}$, the authors estimate:

$$X_{i,t} = \alpha_i + \beta_1 \text{PostSample}_{i,t} + \beta_2 \text{PostPublication}_{i,t} + e_{i,t}. \quad (6)$$

Interpretation: Post-publication increases in trading volume, dollar volume, and short interest are direct evidence that publication draws investor attention and trading activity toward anomaly strategies.

3.9 Comovement with other published predictors

To study changes in return correlations, the paper estimates regressions of each predictor's return on indices of other predictors:

$$R_{i,t} = \alpha_i + \beta_1 \text{Index}_{i,t}^{\text{in}} + \beta_2 \text{Index}_{i,t}^{\text{pub}} + \gamma_1(\text{PostPublication}_{i,t} \times \text{Index}_{i,t}^{\text{in}}) + \gamma_2(\text{PostPublication}_{i,t} \times \text{Index}_{i,t}^{\text{pub}}) + \delta \text{PostPublication}_{i,t} + e_{i,t} \quad (7)$$

Interpretation: Publication changes the return comovement structure. Published predictors begin to move more with other published predictors, which is consistent with arbitrageurs trading multiple published anomalies together.

4 Identification and empirical design

4.1 What identifies data mining versus publication effects

- The original sample period is chosen before the authors' out-of-sample period.
- If published predictors were mostly data-mined, returns should fall immediately after the original sample ends.
- Publication is a separate event that potentially changes investor information.
- Comparing post-sample pre-publication returns with post-publication returns separates statistical bias from publication-related trading.

Interpretation: The design is not a randomized experiment, but it uses the timing of original sample ends and publication dates to decompose return decay.

4.2 Why the post-sample coefficient is an upper bound on statistical bias

- Some sophisticated traders may know about a predictor before formal publication.
- Authors, seminar audiences, referees, SSRN readers, and industry contacts can transmit information earlier.
- If early trading occurs before publication, the post-sample period already contains some arbitrage effect.

Interpretation: Because of early learning, the out-of-sample decline overstates pure data mining. Therefore, the remaining post-publication effect is a lower bound on publication-informed trading.

4.3 Why the post-publication coefficient is informative

- Publication makes the predictor easier to discover and implement.
- If predictability reflects rational risk compensation, publication should not cause a large additional decline.
- If predictability reflects mispricing, publication can attract arbitrage capital and lower expected returns.

Interpretation: The large additional decline after publication supports the mispricing-and-learning interpretation.

4.4 Standard errors and cross-predictor dependence

- Predictor returns are not independent because many anomalies trade overlapping stocks.
- The authors use feasible generalized least squares under contemporaneous cross-correlation of predictor returns.
- Clustering by time gives similar but slightly smaller standard errors.

Interpretation: The inference accounts for the fact that multiple predictor portfolios can be exposed to common market-wide or anomaly-wide shocks.

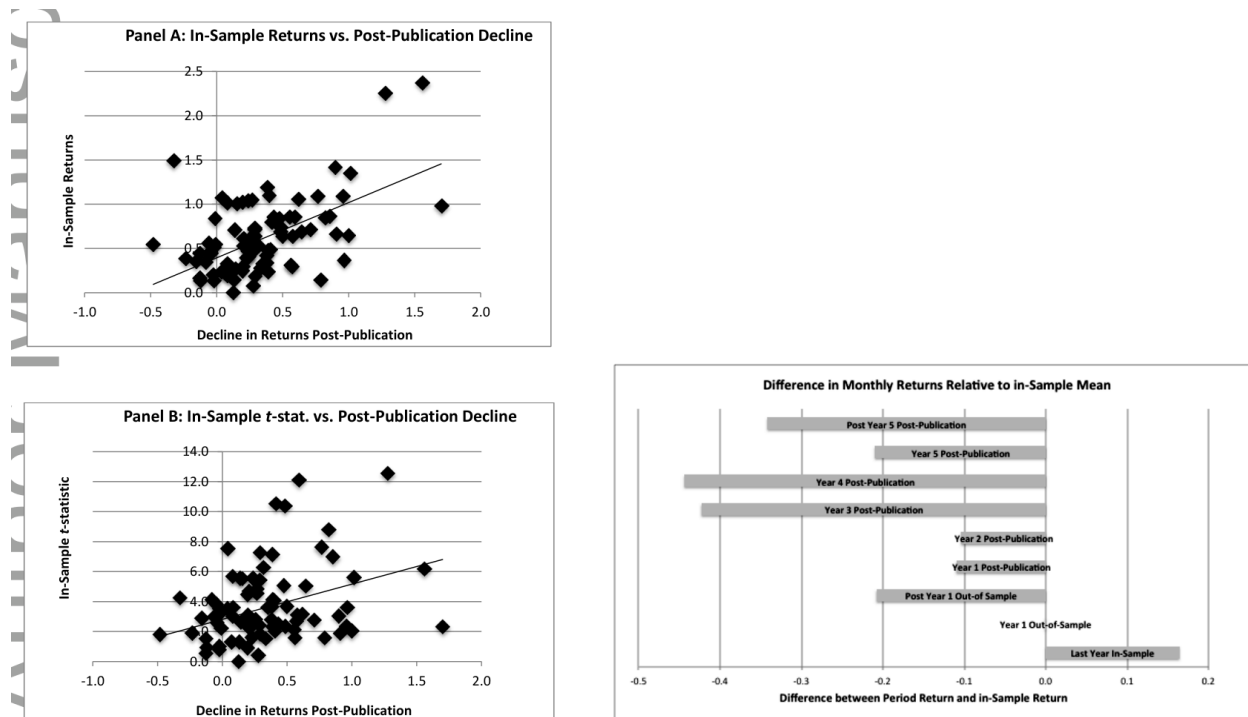
4.5 Main limitations

- Publication date is an imperfect proxy for when investors learn.
- Some predictors cannot be exactly reconstructed from the original papers.
- The sample includes only published predictors, not the universe of tested but unpublished predictors.
- The decomposition between data mining and publication effects is bounded rather than exact.
- The paper documents reduced returns and trading activity but does not observe investor learning directly.

Interpretation: The evidence is strongest as a broad cross-predictor pattern: many anomalies decay after publication, but the data cannot pin down exact investor learning dates.

5 Tables and figures

5.1 Figure 1 + Figure 2: strongest predictors decay more and returns decline around sample/publication dates



(a) Figure 1: in-sample strength versus post-publication decline.

(b) Figure 2: predictor return dynamics around sample end and publication.

Read:

- Figure 1, Panel A shows a positive relation between in-sample returns and post-publication declines.
- Figure 1, Panel B shows a similar relation using in-sample t -statistics.
- Figure 2 shows that the last in-sample year is positive, but returns drop in the out-of-sample and post-publication periods.
- The largest declines occur several years after publication, consistent with gradual diffusion of the anomaly and gradual arbitrage.

Economic message: Stronger published results attract more attention. The market reacts more to the anomalies that look most profitable.

5.2 Table I + Table II: summary statistics and baseline return decay

This table reports summary statistics for the predictors studied in this paper. The returns are equal-weighted predictor portfolio, that is, we first estimate the statistic for each predictor portfolio, and then take an equal-weighted average across predictors portfolio. The reported standard deviations are the standard deviations of the predictors' mean returns. Our sample period ends in 2013.

Number of Predictor Portfolios	97
Predictors Portfolios with t -statistic > 1.5	83 (86%)
Mean Publication Year	2000
Median Publication Year	2001
Predictors from Finance journals	68 (70%)
Predictors from Accounting journals	27 (28%)
Predictors from Economics journals	2 (2%)
Mean Portfolio Return In-Sample	0.582
Standard Deviation of Mean In-Sample Portfolio Return	0.395
Mean Observations In-Sample	323
Mean Portfolio Return Out-of-Sample	0.402
Std. Dev. of Mean Out-of-Sample Portfolio Return	0.651
Mean Observations Out-of-Sample	56
Mean Portfolio Return Post-Publication	0.264
Std. Dev. of Mean Post-Publication Portfolio Return	0.516
Mean Observations Post-Publication	156

(a) Table I: summary statistics.

Table II
Regression of predictor portfolio returns on post-sample and post-publication indicators

The regressions test for changes in returns relative to the predictor's sample-end and publication dates. The dependent variable is the monthly return to a long-short portfolio that is based on the extreme quintiles of each predictor. Post-Sample (S) is equal to one if the month is after the sample period used in the original study and zero otherwise. Post-Publication (P) is equal to one if the month is after the official publication date and zero otherwise. Mean is the in-sample mean return of the predictor portfolio during the original sample period. t -statistics are the in-sample t -statistic of each predictor portfolio. Standard errors (in parentheses) are computed under the assumption of contemporaneous cross-sectional correlation between panel portfolio residuals. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels respectively. The bottom three rows report p -values from tests of whether post-sample and post-publication changes in returns are statistically different from one another and whether any declines are 100% of the in-sample mean (the effects disappears entirely).

Variables	(1)	(2)	(3)	(4)
Post-Sample (S)	-0.150*** (0.077)	-0.180** (0.085)	0.157 (0.103)	0.067 (0.112)
Post-Pub. (P)	-0.337*** (0.090)	-0.387*** (0.097)	-0.002 (0.078)	-0.120 (0.114)
S x Mean			-0.532*** (0.221)	
P x Mean			-0.548*** (0.178)	
S x t -statistic				-0.061*** (0.023)
P x t -statistic				-0.063*** (0.018)
Predictor FE?	Yes	Yes	Yes	Yes
Observations	51,851	45,465	51,851	51,944
Predictors (N)	97	85	97	97
Null : S=P	0.024	0.021	NA	NA
Null: P=1*(Mean)	0.000	0.000		
Null: S=1*(Mean)	0.000	0.000		

(b) Table II: post-sample and post-publication indicators.

Read:

- Table I reports 97 predictor portfolios and 83 portfolios with in-sample t -statistics above 1.5.
- Mean in-sample return is 0.582%, mean out-of-sample return is 0.402%, and mean post-publication return is 0.264%.
- Table II estimates a post-sample coefficient of -0.150 and a post-publication coefficient of -0.337 in the baseline specification.
- Relative to the 0.582% in-sample average, the post-sample decline is about 26% and the post-publication decline is about 58%.
- The bottom rows reject both no post-publication decay and complete disappearance of post-publication returns.

Economic message: The main result is simple and strong: anomalies lose substantial return predictability after publication, but do not vanish completely.

5.3 Table III + Table IV: time trends, persistence, and predictor types

Table III
Time Trends and Persistence in Predictor Returns

The regressions reported in this table test for time trends and persistence in predictor returns. Post-Sample (S) is equal to one if the month is after the sample period used in the original study and zero otherwise. Post-Publication (P) is equal to one if the month is after the official publication date and zero otherwise. Time is the number of months divided by 100 post-January 1926. Post-1993 is equal to one if the year is greater than 1993 and zero otherwise. All indicator variables are equal to zero if they are not equal to one. 1-Month Return and 12-Month Return are the predictor's return from the last month and the cumulative return over the last twelve months. Standard errors (in parentheses) are computed under the assumption of contemporaneous cross-sectional correlation between panel portfolio residuals. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Time	-0.069*** (0.011)		-0.069*** (0.026)			
Post-1993		-0.120 (0.074)	0.303*** (0.118)			
Post-sample			-0.190** (0.081)	-0.179** (0.080)	-0.132* (0.076)	-0.128 (0.078)
Post Pub.			-0.362*** (0.124)	-0.310** (0.122)	-0.295*** (0.089)	-0.258*** (0.093)
1-Month Return					0.114*** (0.015)	
12-Month Return						0.020*** (0.004)
Observations	51,851	51,851	51,851	51,851	51,754	50,687
Char. FE?	Yes	Yes	Yes	Yes	Yes	Yes
Time FE?	No	No	No	Yes	No	No

(a) Table III: time trends and persistence.

Table IV
(Continued)

VARIABLE	(1)	(2)	(3)	(4)
Post-Publication (P)	-0.208*** (0.059)	-0.316*** (0.097)	-0.310*** (0.080)	-0.301*** (0.089)
Market	0.304*** (0.079)			
P x Market	-0.244 (0.169)			
Event		-0.098** (0.046)		
P x Event		0.105 (0.091)		
Valuation			-0.056 (0.063)	
P x Valuation			0.186* (0.131)	
Fundamental				-0.201*** (0.045)
P x Fundamental				0.025 (0.089)
Constant	0.482*** (0.036)	0.606*** (0.052)	0.585*** (0.000)	0.630*** (0.053)
Observations	51,851	51,851	51,851	51,851
Predictors	97	97	97	97
Type + (P x Type)	0.060	0.007	0.121	-0.176
p-value	0.210	0.922	0.256	0.012

(b) Table IV: predictor returns by type.

Read:

- Table III shows that a negative post-publication coefficient remains after adding a time trend, a post-1993 dummy, time fixed effects, and lagged return controls.
- Lagged one-month and twelve-month returns are positive, indicating some return persistence.
- Table IV separates market, event, valuation, and fundamental predictors.
- Post-publication decay is negative across all predictor types.
- Market-based predictors have a large baseline premium and then decline after publication.

Economic message: The return decay is not merely a secular trend or a simple persistence effect. It is tied to publication timing and appears across different categories of predictors.

5.4 Table V + Table VI: arbitrage costs and trading activity

Read:

- Table V relates post-publication returns to size, spreads, dollar volume, idiosyncratic risk, dividends, and a costly-arbitrage index.
- Returns persist more when arbitrage is costly, supporting a limits-to-arbitrage interpretation.
- Table VI shows that trading volume, dollar volume, and short interest increase after publication.
- The increase in short interest is especially consistent with arbitrageurs trading the short side of anomalies.

Economic message: Publication changes behavior, not just statistical estimates. Investors appear to trade more aggressively on published predictors, especially where arbitrage is feasible.

Table V
(Continued)

Variables	(1)	(2)	(3)	(4)	(5)	(6)
Post-Pub. (P)	-0.190 (0.274)	-0.139** (0.235)	0.215*** (0.230)	-0.242 (0.273)	-0.321** (0.211)	-0.264*** (0.001)
P x Size	-0.138 (0.459)					
Size	-1.064** (0.236)					
P x Spreads		-0.301 (0.603)				
Spreads		1.228** (0.252)				
P x Dol. Vol.			-1.059** (0.500)			
Dol. Vol.			0.215 (0.308)			
P x Idio. Risk				-0.047 (0.554)		
Idio. Risk				2.064*** (0.330)		
P x Div.				-0.321 (0.211)		
Div.				-0.526*** (0.145)		
P x Index					-0.009 (0.019)	
Index					-0.056*** (0.011)	
Constant	1.145*** (0.130)	0.146* (0.174)	0.476*** (0.144)	-0.469*** (0.171)	0.855*** (0.097)	0.565*** (0.000)
Observations	51,851	51,851	51,851	51,851	51,851	51,851
CA + (P x CA)	-1.202	0.927	-0.844	2.017	-0.847	-0.065
p-value	0.003	0.096	0.000	0.000	0.144	0.000

Table VI
Trading activity dynamics in predictor portfolio stocks

This regression models the dynamics of the traits of stocks in predictor portfolios, relative to the predictor's original sample period and the publication date. We perform monthly ranks based on turnover, dollar value of trading volume, and stock return variance. Trading Volume is measured as shares traded, while Dollar Volume is measured as shares traded multiplied by price. Variance is the monthly stock return squared. For each predictor portfolio, we compute the average of each variable among the stocks that enter either the long or the short side of the characteristic portfolio each month, and test whether it increases out-of-sample and post-publication. For short interest (shares shorted scaled by shares outstanding), we take the average short interest in the short quintile for each characteristic, and subtract from it the average short interest in the long quintile. The short interest findings are reported in percent (the dependent variable is multiplied by 100). *Post-sample* is equal to 1 if the month is after the end of the sample but pre-publication. *Post-Sample (S)* is equal to one if the month is after the sample period used in the original study and zero otherwise. *Post-Publication (P)* is equal to one if the month is after the official publication date and zero otherwise. Standard errors (in parentheses) are computed under the assumption of contemporaneous cross-sectional correlation between panel portfolio residuals. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Variables	Variance	Trading Volume	Dollar Volume	Short - Long Short Interest
Post-Sample (S)	-0.054*** (0.007)	0.092*** (0.001)	0.066*** (0.007)	0.166*** (0.014)
Post-Pub. (P)	-0.065*** (0.008)	0.187*** (0.013)	0.097*** (0.007)	0.315*** (0.013)
Observations	52,632	52,632	52,632	41,026
Time FE?	Yes	Yes	Yes	No
Predictor FE?	Yes	Yes	Yes	Yes
Null: S=P	0.156	0.000	0.000	0.000

5.5 Table VII: comovement with other predictor indices

Table VII
Regressions of predictor returns on return indices of other predictors

This regression models the returns of each predictor relative to the returns of other predictors. The dependent variable is a predictor's monthly long-short return. *Post-Publication (P)* is equal to one if the month is a official publication date and zero otherwise. *In-Sample Index Return* is the equal-weighted return of all unpublished predictor portfolios. *Post-Publication Index Return* is an equal-weighted return of all published predictor portfolios. Standard errors (in parentheses) are computed under the assumption of contemporaneous cross-sectional correlation between panel portfolio residuals. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Variables	Coefficients
In-Sample Index Returns	0.748*** (0.000)
Post-Publication Index Return	-0.008 (0.243)
P x In-Sample Index Returns	-0.674*** (0.425)
P x Post-Publication Index Return	0.652*** (0.576)
Publication (P)	-0.880 (0.544)
Constant	0.144*** (0.267)
Observations	42,975
Predictors	97

Figure 4: Table VII: regressions on return indices of other predictors.

Read:

- In-sample index returns have a positive coefficient of 0.748.
- The interaction between publication and in-sample index returns is negative.
- The interaction between publication and the post-publication index return is positive, 0.652.

- The result implies that a predictor becomes more correlated with other already published predictors after publication.

Economic message: Published anomalies start to move together, consistent with arbitrageurs learning about and trading multiple published strategies.

6 Short summary

- The paper asks whether academic research destroys stock return predictability.
- It studies 97 published cross-sectional predictors from 80 papers.
- Each predictor is converted into a monthly long-short portfolio return.
- The paper compares in-sample, post-sample pre-publication, and post-publication periods.
- Mean returns fall from 0.582% in sample to 0.402% out of sample and 0.264% post-publication.
- The post-sample decline is about 26%, interpreted as an upper bound on statistical bias.
- The post-publication decline is about 58%, implying a lower-bound publication effect of about 32%.
- Predictors with stronger in-sample results decline more after publication.
- The decline is robust to time trends, a post-1993 dummy, time fixed effects, and persistence controls.
- Returns persist more when arbitrage is difficult.
- Trading activity rises after publication.
- Predictor returns become more correlated with other published predictors after publication.
- The evidence supports a mixed view: some predictability is statistical bias, some reflects mispricing that investors learn about, and some returns persist because arbitrage is costly.

7 Conclusion

7.1 Core conclusion

McLean and Pontiff show that published stock-return predictors decay substantially after the original sample ends and decay even more after publication. The evidence suggests that both statistical bias and publication-informed arbitrage matter. Academic research does not fully eliminate predictability, but it reduces it meaningfully.

7.2 Economic intuition

The intuition is that publication turns a private or obscure trading signal into public knowledge. If the signal was partly a data-mined artifact, it fails out of sample. If the signal reflected mispricing, investors trade on it after publication and push prices closer to efficiency. If arbitrage is costly, some predictability remains.

7.3 Takeaway

The paper's main takeaway is not that anomalies are fake or that markets instantly become efficient. The takeaway is that research changes markets. Published predictors are weaker after publication because investors learn from academic work and arbitrage away part of the opportunity.